Name_____ Date_____ Period_____

LESSON 3-3 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Equivalent Ratios and Tables

Maryland Standard 6.RP.A.3a · Learning target – I can build a ratio table by scaling both rows by the same factor.

REMEMBER

- **Equivalent ratios** make the same comparison. Multiply or divide **both** terms by the same number.
- A **ratio table** lists equivalent ratios. Each column is one pair.
- Adding the same number to both terms does **not** keep the ratio.

WORKED EXAMPLE

Lemonade uses 3 lemons for every 4 cups of water. How many cups of water go with 12 lemons?

Step 1 Find the factor: 3 × 4 = 12

Step 2 Multiply the other term by the same factor: 4 × 4 = 16

Answer **16 cups of water**

WORD BANK

ratio – compares two amounts, like 2 : 5

equivalent – has the same value

ratio table – a table of equal ratios; each column is one pair

factor – the number you multiply by

PRACTICE

show your work · label units

1. Trail mix uses 2 cups of nuts for every 5 cups of cereal. Complete the ratio table.

Step 1 Find what 2 was multiplied by to get each top number.

Step 2 Multiply 5 by that same number.

Nuts (cups)	2	4		10	
Cereal (cups)	5		15		35

2. Fill in the missing number so the ratios are equivalent.

Step 1 Find the factor that links the two numbers you know.

Step 2 Use the same factor on the other term.

a. 3 : 4 = 15 : ____

b. 6 : 7 = ____ : 42

c. 18 : 24 = 3 : ____

d. 9 : 2 = 63 : ____

3. Select all the ratios that are equivalent to 4 : 6.

Step 1 Divide 4 : 6 by 2. Is that ratio a choice?

Step 2 Multiply 2 : 3 by 2, by 3, by 4, by 5. Look for those ratios.

Step 3 For each other choice, ask: was it multiplied or added?

□ **A** 2 : 3 □ **B** 8 : 12 □ **C** 6 : 8 □ **D** 10 : 15

4. **Bracelets.** Nadia uses **4 gold** beads for every **9 purple** beads. She uses **36 purple** beads. How many **gold** beads does she use? Show a ratio table.

Step 1 Find the factor: 9 × ___ = 36.

Step 2 Multiply 4 by the same factor.

Show your work.

Answer_____ gold beads

5. **Paint Mix.** A paint color uses 3 cups of red for every 2 cups of white. Lou uses 21 cups of red.

Step 1 Find the factor: 3 × ___ = 21.

Step 2 Multiply 2 by the same factor.

Step 3 Add red and white to find the total.

a. How many cups of white does Lou use? _____ cups

b. How many cups of paint does Lou make in all? _____ cups

6. **Find the Error.** Mia says 5 : 8 is equivalent to 8 : 11 because she added 3 to each term. Is she right? Explain and give a ratio that is equivalent to 5 : 8.

Frame *Mia is ___ because adding 3 ___. A ratio equivalent to 5 : 8 is ___.*

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

AT HOME Double a family recipe out loud. Which amounts change, and by what factor?

Name_____ Date_____ Period_____

LESSON 3-3 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

The Bridge Strategy (Scaling Ratios)

Maryland Standard 6.RP.A.3a · Learning target – I can use the bridge strategy to scale a ratio when I cannot multiply by a whole number.

REMEMBER

- No whole number gets you to the target? **Build a bridge.**
- **Divide** both terms down to a smaller equivalent ratio, then **multiply** both terms up to the target.

WORKED EXAMPLE

A drink uses 6 scoops of mix for 8 cups of water. How much water goes with 9 scoops?

Step 1 $6 \times ? = 9$ is not a whole number.
Bridge down: $6 : 8 \div 2 = 3 : 4$

Step 2 Scale up: $3 : 4 \times 3 = 9 : 12$

Answer **12 cups of water**

WORD BANK

bridge ratio – a smaller ratio you make first

simplify – divide both numbers by the same number

target – the number you need to reach

PRACTICE

show your work · label units

1. A machine fills 15 bottles every 6 minutes. Use the bridge column to find the bottles for 10 minutes.

Step 1 $6 \div 3 = 2$. Divide 15 by 3 too.

Step 2 Then multiply the 2-minute column by 5.

Minutes	6	2	10
Bottles	15		

2. Use a bridge to find each missing number.

a. $4 : 6 = 10 :$ ____

b. $9 : 12 = 6 :$ ____

3. **Map Scale.** On a map, **4 cm** stand for **30 km**. How many **km** do **10 cm** stand for?

Step 1 Divide both numbers by 2 to make the bridge.

Step 2 Multiply both bridge numbers by 5.

a. Bridge ratio: _____

b. Kilometers for 10 cm: _____ km

4. **Bread.** A recipe uses 10 cups of flour for 4 loaves of bread.

Step 1 Divide 10 and 4 by 2.

Step 2 Multiply the bridge by 3 to reach 6 loaves.

a. Bridge ratio: _____

b. Cups of flour for 6 loaves: _____ cups

5. **Find the Error.** A glaze uses 6 spoons honey to 10 spoons water. For 15 spoons honey, Ana added 9 to both terms: 15 : 19. Explain her mistake and fix it.

Frame *Ana's mistake is ___. The correct amount of water is ___ because ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. The ratio of cups of rice to cups of water is 12 : 8. How many cups of water go with 15 cups of rice?

A 10 **B** 11 **C** 12 **D** 22.5

7. Which steps scale 6 : 9 to 10 : ___ using the bridge strategy?

A ÷ 3, then × 5 **B** + 4 **C** × 2, then − 2 **D** ÷ 6, then × 9

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

AT HOME Explain to someone at home why you divide first when you cannot multiply straight to the number you need.